Random Number

OTP

$|m| = l$

$|k| = l$

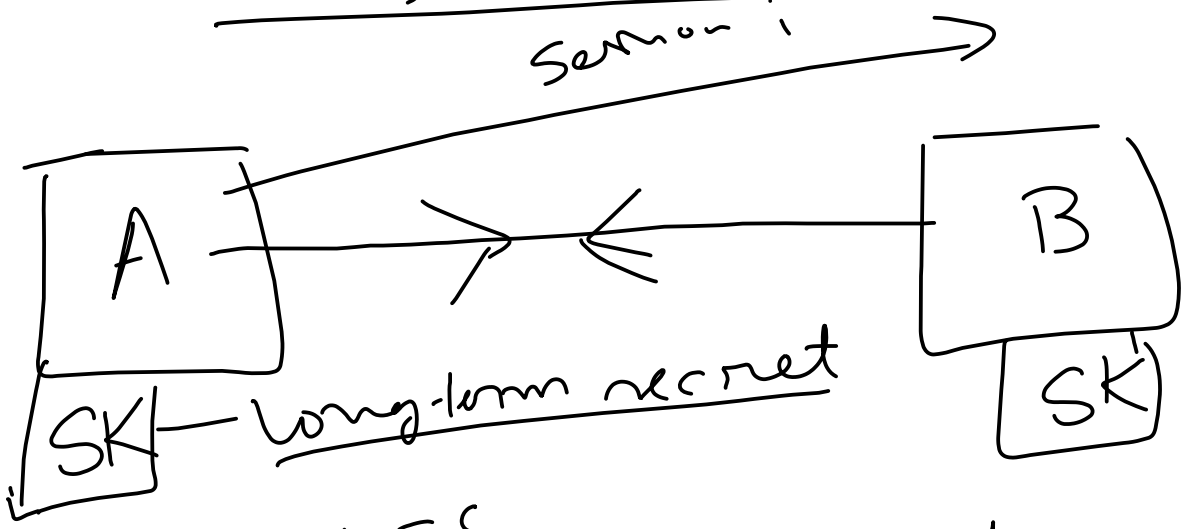
$m_1 \oplus k = c_1$

$m_2 \oplus k = c_2$

$c = c_1 \oplus c_2 = m_1 \oplus m_2$



Symmetric Key



$m \oplus SK = c$

$c \oplus SK = m$

composed

composed

$KDF(SK, i) =$ session key

Set session i

PRF $i = 0, 1, 2, \ldots$

$S = \{0, 1, 2, 3, 4\}$

$P(X = i) \quad i = 0(1)4$

$= \frac{1}{5}$

X is a random variable which can take values

0, 1, 2, 3, 4

One unbiased coin

H = 1 p 1/2

T = 0 p 1/2

H = 1 p

T = 0 (1-p)

biased coin $\frac{3}{4}$

1 3 0 4 ? (equal)

1 1 1 1

$\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$ $\frac{1}{5}$

1 2 3 4

H T T H

H - 1/4

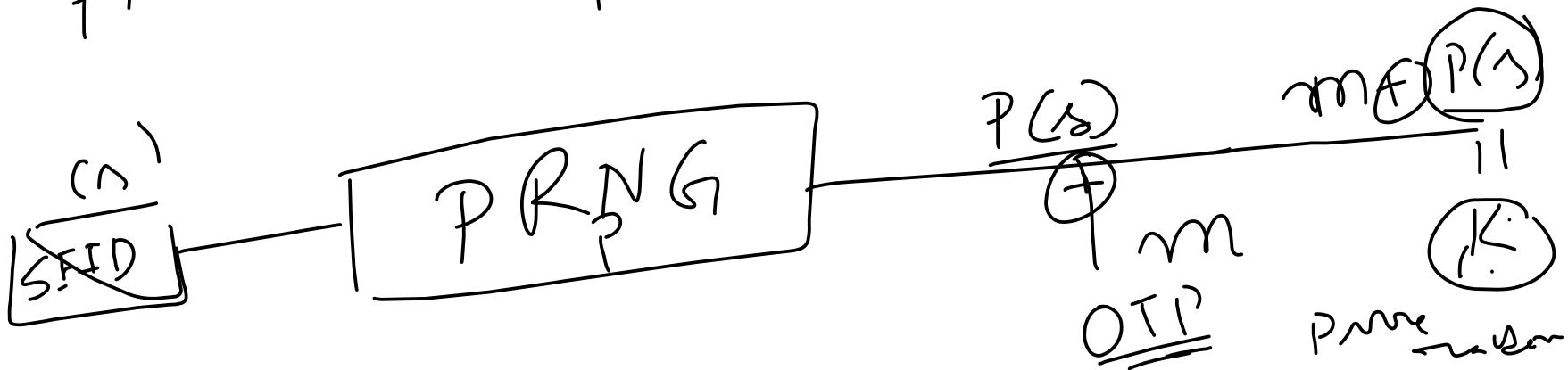
T - 3/4



Pseudorandom Number

Look like pure random number.
Generated by a deterministic algorithm
— Digital computer

Should look like random so that
any polytime adversary cannot distinguish
between a PRN and pure random number.



$$X_{n+1} = (aX_n + c) \bmod m$$

$$U_n = \frac{X_n}{m} \quad [0, 1)$$

X_0 = seed
a = multiplier
c = increment
m = modulus

$X_n \in \{0, 1, \ldots, m\}$

$m = 16$
$a = 5$
$c = 3$

$X_0 = 1$

$X_1 = (5 \cdot 1 + 3) \bmod 16 = 8$

$X_2 = (5 \cdot 8 + 3) \bmod 16 = 11$

$X_3 = (5 \cdot 11 + 3) \bmod 16 = 10$

$X_4 = (5 \cdot 10 + 3) \bmod 16 = 5$

$X_5 = (5 \cdot 5 + 3) \bmod 16 = 12$

Period (12)

{1, 8, 11, 10, 5, 12, 15, 14, 9, 0, 3, 6, 1}

Period of LCG

The maximum period of m is achieved

1. gcd (c, m) = 1
2. (a - 1) is divisible by prim factors of m
3. If m is divisible by 4, (a - 1) is also divisible by 4

16 2

Hull Dobell condition.

$p \equiv 3 \bmod 4$

$q \equiv 3 \bmod 4$

$a_{i+1} = a_i^2 \bmod n$

$a_0 = 10$

$a_1 = 10^2 / 77 = 23 - 1$

$a_2 = 23^2 / 77 = 67 - 1$

$a_3 = 67^2 / 77 = 2^3 - 1$

| | | | | | | |

$n = pq$

$a_0 \in Z_n^*$

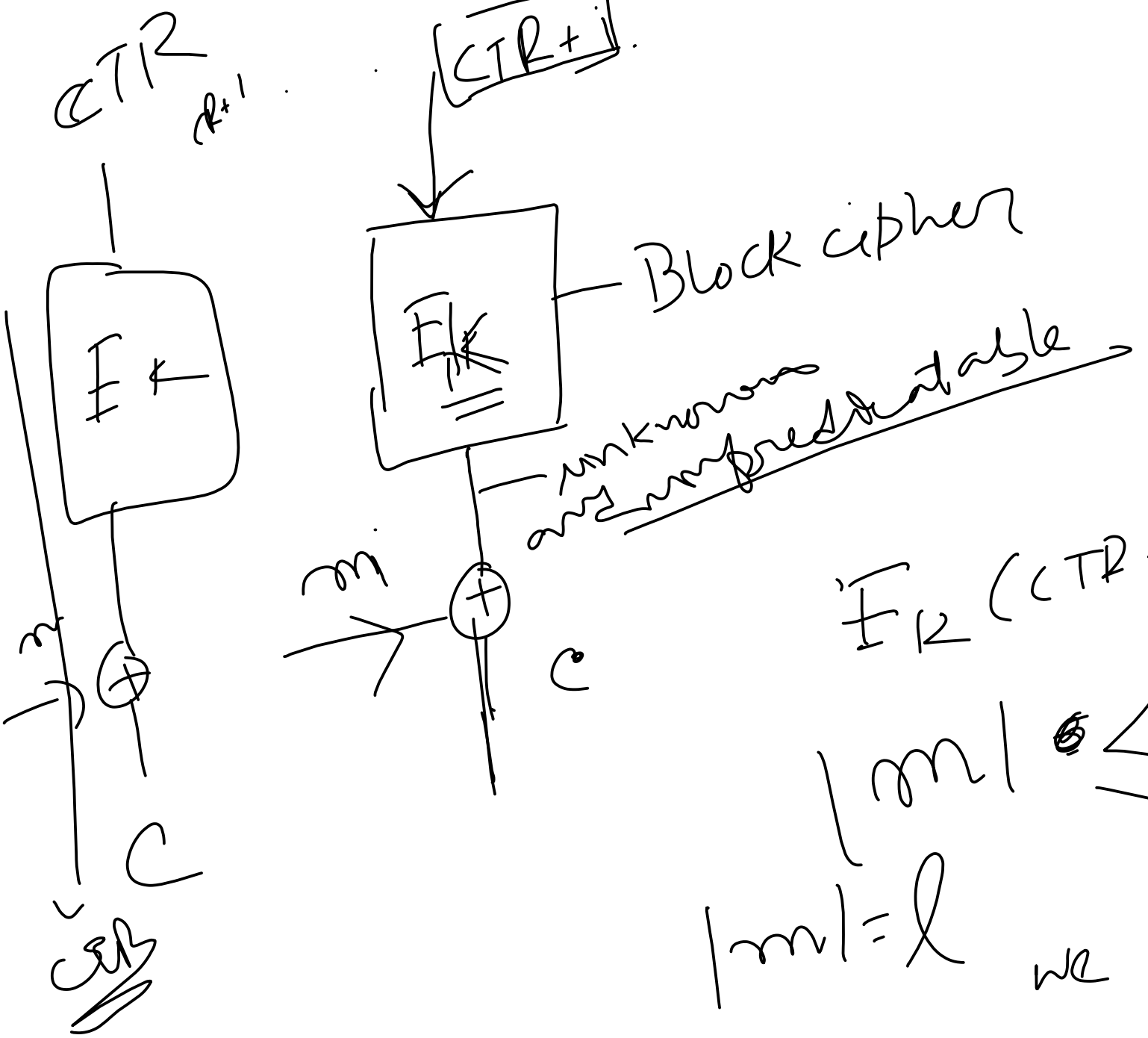
$p = 7$ $\quad Z_n / \{0\}$

$q = 11$

$n = 77$

Security is based on the hardness of factoring

	LCG	BBS
Linear Structure	YES $X_{i+1} = (aX_i + b) \bmod m$	NO $X_{i+1} = X_i^2 \bmod n$ $n = p, q$
CRYPTO GRAPHICALLY SECURE	NO	YES
SPEED	VERY FAST	SLOW
SECURITY BIASES	HURFSTIC	factoring large prime
	SIMULATION	RARE (slow)



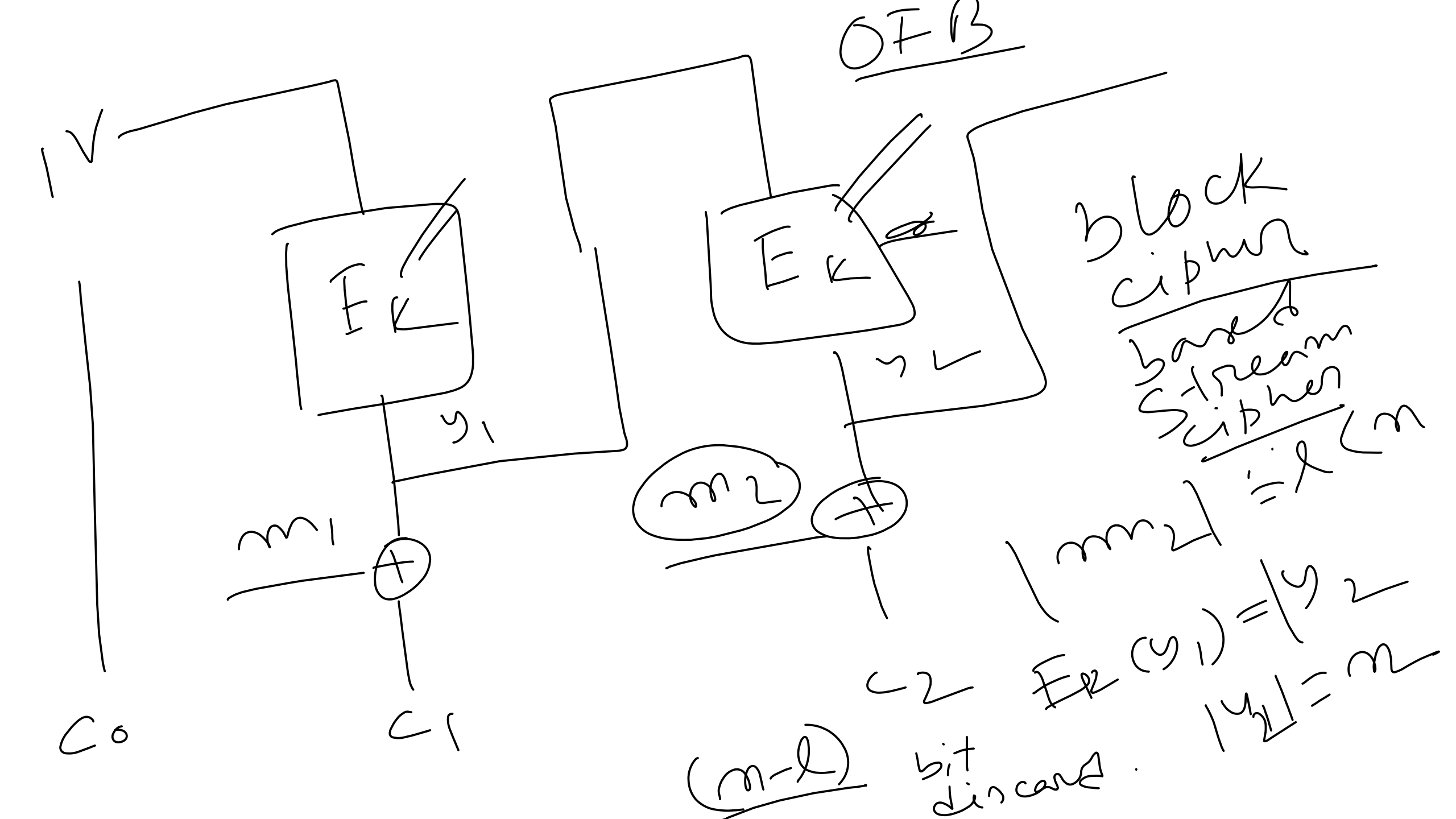
CTR counter

$r \in \{0,1\}^n$

$F_K(CTR+i) = y_i \in \{0,1\}^n$

$|m| \leq n$ y' from y_i as per need.

$|m| = l$ we discard $(n-l)$ bits


OFB
IV
EK
EK
y1
y2
m1
m2
c0
c1
c2
block cipher
based
stream
cipher
|m| = k
|m2|
EK(y1) = |y2
|y2| = m
(m-l) bit
discard